RESQNET AI (VERSION 2.0)

Technical Architecture, False-Positive Filtering, and Handbook Compliance Specification

Problem Statement: 1.3 RoadSoS | **Host:** Centre of Excellence for Road Safety (CoERS), IIT Madras

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1. Executive Summary & System Evolution

ResQNet AI v2.0 represents an upgraded, production-grade architecture designed specifically to align with the core constraints of the IIT Madras RoadSoS problem statement. Beyond establishing an offline-first mobile AI assistant, this version integrates deep signal filtering layers to distinguish real-world road noise from actual life-threatening vehicular accidents. It establishes a multi-stage validation engine that functions completely over local computing nodes, preventing false positive alarms without sacrificing safety during moments of actual user incapacitation.

2. Technical Architecture & Component Rationale

Following the handbook mandate favoring open-source stacks, free APIs, and local processing pipelines, the system components have been selected to ensure full zero-network compliance:

Layer Block	Technology Selected	Strategic Engineering Purpose
Computational Engine	`Python 3.11` / `ONNX Runtime`	Preferred by IITM guidelines. Enables lightning-fast inference of quantized mathematical operations.
On-Device Storage	`DuckDB` + `HNSW Extensions`	In-process relational store capable of holding millions of pre-loaded geo-coordinates with indexed query execution.
Speech Processing	`Vosk Acoustic Engine`	Fully offline automatic speech recognition engine operating locally within restricted memory boundaries (< 50MB RAM).
Peer-to-Peer Mesh	`BLE 5.2 (Bluetooth Low Energy)`	Enables hardware beacons to broadcast asynchronous rescue packets in complete cellular dead-zones.

3. False-Positive Mitigation & Two-Stage Validation Pipeline

A critical challenge in mobile telematics is distinguishing standard environmental variations (such as highly damaged roads, continuous rough braking, speed bumps, and device drops) from actual vehicular collision profiles. ResQNet AI prevents user annoyance and false emergency dispatches via a robust, hardware-software collaborative validation pipeline.

3.1 Stage 1: Mathematical Sensor Fusion & Threshold Filtering

The device monitors spatial acceleration vectors continually at high sampling rates (50 Hz). To distinguish random shaking from structural impact, the system processes the raw telemetry through a localized multi-axis filter.

Mathematical Evaluation Matrix:

The engine continually tracks the global Vector Magnitude Acceleration (A_{mag}):

$$A_{mag} = \sqrt{(A_x^2 + A_y^2 + A_z^2)}$$

- **Rough Roads / Heavy Braking:** Generates erratic high-frequency vibrations but limits $A_{mag} \leq 1.8G$ *to* $2.2G$. The gyroscopic structural orientation remains within normal horizontal planes. The engine automatically filters and drops these instances.
- **Severe Vehicular Crash:** Triggers a sudden, catastrophic deceleration signature where $A_{mag} \geq 4.5G$ within a strict temporal frame of $\Delta t \leq 120ms$, accompanied by an abrupt rotational angular tilt exceeding $60°/sec$. This instantly triggers the Stage 2 verification window.

3.2 Stage 2: Voice-First Persistence Verification Loop

If the mathematical thresholds of Stage 1 are breached, the application transitions into an interactive safety appraisal sequence to ensure the user is not disturbed by accidental false alerts:

1. **The Silent Voice Window (15 Seconds):** Instead of launching deafening sirens immediately, the application activates the local device microphone and uses the offline Vosk engine to play a clear voice prompt: *"Potential impact detected. Are you safe? Please state your status."*
2. **Continuous Natural Processing Loop:** The offline ASR engine actively listens for specific local reset phrases (e.g., *"I am safe", "I am fine", "Cancel alarm", "Main theek hu"*). If a positive verbal confirmation is parsed within this 15-second window, the system immediately cancels the alert, flushes sensor memory buffers, and resets safely without calling emergency nodes.
3. **The Unresponsive Siren Escalation (10 Seconds):** If the system receives absolute silence or noise that does not translate to safety markers within the 15-second window (implying the driver is incapacitated, severely injured, or unconscious), the system escalates. It activates a high-intensity, maximum-volume local haptic siren and physical vibration for 10 seconds. This secondary alert serves to capture the attention of aypassers or give the user one last physical window to click a large "Dismiss" element on screen.

4. **Autonomous Execution:** If both the 15-second voice validation window and the 10-second loud alarm pass with zero user response, the system declares a definitive catastrophic emergency and switches directly into autonomous rescue execution.

4. Handbook Compliance & Evaluation Rules Alignment

Per the official rules issued by the Centre of Excellence for Road Safety (CoERS), IIT Madras, submissions must satisfy explicit algorithmic, architectural, and copyright rubrics. ResQNet AI is engineered precisely around these evaluation targets.

4.1 Code Quality, Authenticity & Plagiarism Policies

As detailed in Section 8.2 of the official handbook, the codebase must represent original, authored work. The evaluation committee uses specialized software processing systems (such as MOSS) to cross-verify submissions:

- **Open Source Model Integration:** All code structures integrate localized open weights explicitly (e.g., quantized ONNX graphs). The implementation details are documented transparently in a localized code layout, avoiding black-box third-party API dependencies that violate the offline mandate.
- **Modular Code Standards:** The submission code package must avoid single, unstructured script designs. It is segregated into clean, test-driven modular formats: `sensors.py` (telemetry and Haversine filters), `database.py` (DuckDB vector lookups), and `triage.py` (Vosk speech logic loops).
- **Submission Payload Requirements:** To clear Stage 1 evaluation on the Unstop portal, the final compressed package strictly packages the entire executable codebase, the completely indexed offline database binary files, a comprehensive `README.md`, and a dedicated `Assumptions.docx` log outlining package version specifications.

4.2 Key Grading Metrics Analysis

The IIT Madras jury scores submissions across specific parameters defined in Section 1.3.4. ResQNet AI targets maximum points in each bracket:

Handbook Criteria	ResQNet AI Implementation Mapping
Reliability & Data Accuracy	Uses immutable local DuckDB transactional tables. Eliminates dynamic web-scraping failures or latency during urgent retrieval loops.
Number of Contacts Fetched	Extensive multi-tiered offline index schemas containing regional hospitals, regional trauma wards, breakdown vehicles, and tire mechanic nodes sorted via geofenced grids.
Offline Functionality	100% processing handled locally at the device edge. Zero network calls required for speech processing, threshold tracking, or distance calculation.
Global Applicability	Sim-based Mobile Country Code (MCC) extraction layer. Dynamically adjusts dialing numbers (112 vs 911) and loads country-specific data profiles without any structural modification to the core codebase.

5. Deployment Workflow Execution Diagram

The functional state machine below highlights the deterministic logic executing across the system during operational use, showing the separation between normal micro-vibrations and true emergency triggers:

```
[Continuous Hardware Polling: Acceleration & Gyro Vector Streaming]
                         │
                         ▼
          [Is Amag >= 4.5G & Deceleration Temporal Frame <= 120ms?]
                ├──▶ NO ──▶ [Drop Noise Event: Rough Road/Pothole Filter]
                └──▶ YES ──▶ [Launch 15-Second Silent Local Microphone Window]
                                      │
                ┌─────────────────────┴──────────────────┐
                ▼ ▼
    [Speech Parsed: "I am safe"] [Absolute Silence / No Match]
                │ │
                ▼ ▼
    [Safe Dismiss & Buffers Reset] [Trigger 10-Second High-Volume Siren]
                                                   │
                ┌──────────────────────────────────┴────────────────────┐
                ▼ ▼
    [Manual Screen Tap: Dismiss] [Zero Feedback / User Unconscious]
                │ │
                ▼ ▼
    [Safe Dismiss & Buffers Reset] [CRITICAL EMERGENCY: Activate Autonomous Engine]
                                                                        │
                                                                        ▼
                                                      [1. Run Local DuckDB Haversine
Vector Sort]
                                                      [2. Initialize BLE Mesh Disaster
Beaconing]
                                                      [3. Auto-Dial SIM Baseband Emergency
Nodes]
```